

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

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اس کے دیکھنے والوں کی سب مشکلیں آسان ہو جائیں گی صبح اس کو دیکھ کر جو کام شروع کیا جائیگا پورا ہوگا اور غیبی طریقوں سے رزق آنے لگے گا!

Question # 4 of 10 (Start time: 03:44:14 PM, 18 August 2022)

By using _____ we avoid the recursive method of traversing a Tree, which makes use of stacks and consumes a lot of memory and time.

Select the correct option



Heap data structure

Email: waqasahmeduog@gmail.com



Threaded binary tree



Binary tree only



Huffman encoding



Question # 5 of 10 (Start time: 03:45:03 PM, 18 August 2022)

For a perfect binary tree of height h , having N nodes, the sum of heights of nodes is _____.

Select the correct option

☒	$N - h - 1$
☐	$N - 1 + h$
☐	$N - (h - 1)$
☐	$N - 1$

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Question # 6 of 10 (Start time: 03:46:32 PM, 18 August 2022)

When a complete binary tree represented by an array then if right child is at position 5 then left child will be at position _____

Select the correct option

☒	4
☐	3
☐	2
☐	6

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Question # 10 of 10 (Start time: 03:48:41 PM, 18 August 2022)

If a tree has 50 nodes, then the total edges/links in the tree will be :

Select the correct option

☐	51
☒	49
☐	55
☐	50

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Question # 1 of 10 (Start time: 03:51:42 PM, 18 August 2022)

See the below code and fill the appropriate answer for ? sign.

```
void fastInorder(TreeNode* p)
{
    while((p=nexInorder(p)) != ? )
        cout << p->getInfo();
}
```

Select the correct option

☐

rootNode

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☐

RTH

☒

dummy

☐

LTH

Question # 1 of 10 (Start time: 03:51:42 PM, 18 August 2022)

See the below code and fill the appropriate answer for ? sign.

```
void fastInorder(TreeNode* p)
{
    while((p=nexInorder(p)) != ? )
        cout << p->getInfo();
}
```

Select the correct option

☐

rootNode

Email: waqasahmeduog@gmail.com

☐

RTH

☒

dummy

☐

LTH

Question # 5 of 10 (Start time: 03:55:09 PM, 18 August 2022)

A complete binary tree of height 3 has between _____ nodes.

Select the correct option

☒	8 to 15
☐	8 to 16
☐	8 to 14
☐	8 to 17

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Question # 6 of 10 (Start time: 03:55:31 PM, 18 August 2022)

The percolateDown procedure will move the smaller value_____ and bigger value_____.

Select the correct option

☐	right,left
☐	left,right
☐	down,up
☒	up,down

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Question # 8 of 10 (Start time: 03:56:01 PM, 18 August 2022)

We are given N items to build a heap of items , this can be done with _____ successive inserts.

Select the correct option

☐	N-1
☐	N+1
☒	N
☐	N^2

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Question # 9 of 10 (Start time: 04:07:58 PM, 18 August 2022)

Which of the following is a property of binary tree?

Select the correct option



A binary tree of N internal nodes has $N+1$ external node.

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A binary tree of N external nodes has $N+1$ internal node.



A binary tree of N external nodes has N internal node.



A binary tree of N internal nodes has $N-1$ external node.

Question # 10 of 10 (Start time: 04:08:40 PM, 18 August 2022)

In complete binary tree the bottom level is filled from -----

Select the correct option



Not filled at all

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Left to right



Right to left



None of the given options

Question # 1 of 10 (Start time: 04:11:14 PM, 18 August 2022)

Consider a min heap, represented by the following array:

3,4,6,7,5,10

After inserting a node with value 1. Which of the following is the updated min heap?

Select the correct option



3,4,6,7,5,10,1



3,4,6,7,5,1,10



1,4,6,7,5,10,3



1,4,3,7,5,10,6

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Question # 4 of 10 (Start time: 04:13:17 PM, 18 August 2022)

Finding the minimum is easy; it is _____ of the min heap.

Select the correct option

☐	Right most child
☐	Left most child
☐	None of the given options.
☒	Top

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Question # 4 of 10 (Start time: 04:13:17 PM, 18 August 2022)

Finding the minimum is easy; it is _____ of the min heap.

Select the correct option

☐

Right most child

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☐

Left most child

☐

None of the given options.

☒

Top

Question # 5 of 10 (Start time: 04:13:51 PM, 18 August 2022)

Which of the following statement is NOT true about threaded binary tree?

Select the correct option



Left thread of the right-most node points to the dummy node.

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The left pointer of dummy node points to the root node of the tree.



Right thread of the right-most node points to the dummy node.



Left thread of the left-most node points to the dummy node.

Question # 6 of 10 (Start time: 04:14:26 PM, 18 August 2022)

By using _____ we avoid the recursive method of traversing a Tree, which makes use of stacks and consumes a lot of memory and time.

Select the correct option



Heap data structure



Threaded binary tree



Huffman encoding



Binary tree only

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Question # 8 of 10 (Start time: 04:16:05 PM, 18 August 2022)

A complete binary tree is a tree that is _____ filled, with the possible exception of the bottom level.

Select the correct option

☒	completely
☐	partly
☐	partially
☐	incompletely

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Question # 9 of 10 (Start time: 04:16:21 PM, 18 August 2022)

Traversing a binary tree can only be done using -----

Select the correct option

☐

Iteration

Email: waqasahmeduog@gmail.com☐

Recursion

☒

Both recursion and iteration

☐

None of the given options

Question # 2 of 10 (Start time: 04:20:18 PM, 18 August 2022)

While building Huffman encoding tree the new node that is the result of joining two nodes has the frequency.

Select the correct option

☐

Equal to the greater frequency

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☐

Equal to the small frequency

☐

Equal to the difference of the two frequencies

☒

Equal to the sum of the two frequencies

Question # 4 of 10 (Start time: 04:22:11 PM, 18 August 2022)

A complete binary tree is a tree that is _____ filled, with the possible exception of the bottom level.

Select the correct option

☐	partially
☐	incompletely
☒	completely
☐	partly

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Question # 5 of 10 (Start time: 04:22:29 PM, 18 August 2022)

We can build a heap in _____ time.

Select the correct option

☐	Polynomial
☒	Linear
☐	Exponential
☐	None of the given options

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Question # 9 of 10 (Start time: 04:24:12 PM, 18 August 2022)

For a perfect binary tree of height h , having N nodes, the sum of heights of nodes is _____.

Select the correct option

 $N - 1$  $N - 1 + h$  $N - h - 1$  $N - (h - 1)$ Email: waqasahmeduog@gmail.com

Question # 3 of 5 (Start time: 04:29:22 PM, 18 August 2022)

A hole is created which needs to be filled if the element is _____ from the heap.

Select the correct option

Deleted

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Inserted



Updated



There is no concept of hole in heap

Question # 3 of 10 (Start time: 04:50:01 PM, 18 August 2022)

If there are 56 internal nodes in a binary tree then how many external nodes this binary tree will have?

Select the correct option

☐	56
☐	55
☒	57
☐	54

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Question # 3 of 10 (Start time: 04:50:01 PM, 18 August 2022)

If there are 56 internal nodes in a binary tree then how many external nodes this binary tree will have?

Select the correct option



56



55



57



54

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Question # 4 of 10 (Start time: 04:50:22 PM, 18 August 2022)

If the height of a perfect binary tree is 4. What will be the total number of nodes in it?

Select the correct option

☐	15
☐	16
☐	32
☒	31

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Question # 4 of 10 (Start time: 04:50:22 PM, 18 August 2022)

If the height of a perfect binary tree is 4. What will be the total number of nodes in it?

Select the correct option

☐	15
☐	16
☐	32
☒	31

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Question # 6 of 10 (Start time: 04:51:08 PM, 18 August 2022)

If both pointers of the node in a binary tree are NULL then it will be a/an

Select the correct option

☐	Inner node
☐	Root node
☐	None of the given options
☒	Leaf node

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Question # 7 of 10 (Start time: 04:51:37 PM, 18 August 2022)

Finding the minimum is easy; it is _____ of the min heap.

Select the correct option

☐

Right most child

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☒

Top

☐

Left most child

☐

None of the given options.

Question # 9 of 10 (Start time: 04:52:05 PM, 18 August 2022)

See the below code and fill the appropriate answer for ? sign.

```
void fastInorder(TreeNode* p)
{
    while((p=nexInorder(p)) != ? )
        cout << p->getInfo();
}
```

Select the correct option

☐

LTH

Email: waqasahmeduog@gmail.com

☐

RTH

☒

dummy

☐

rootNode



Question # 10 of 10 (Start time: 04:52:12 PM, 18 August 2022)

A binary tree with 45 internal nodes has _____links to external nodes.

Select the correct option

☐	44
☒	46
☐	90
☐	45

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Question # 2 of 5 (Start time: 03:32:27 PM, 25 August 2022)

Which of the following formula gives the index position of the left child of "i"?

Select the correct option

☐	$i/2$
☒	$2*i$
☐	$2i+1$
☐	$2+i$

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